

IAAC – Statement of Work (SOW)

1. Title

Professional Services Contractor(s) to Address Stream 1 - Hydrology Knowledge Gaps and Stream 2 - to Identify Areas Most at Risk for Methylmercury Contamination and other Water Quality Knowledge Gaps in the Regional Assessment Area.

2. Objective

The Regional Assessment in the Ring of Fire Area is identifying key information gaps and research needs that could be undertaken to support the Regional Assessment and future impact assessments and decision making, either concurrently or as part of follow-up and monitoring work. Understanding how development and other stressors could impact hydrology and water quality, especially methylmercury contamination in the regional assessment area is a priority for the Regional Assessment Working Group (RAWG). We are looking for a Contractor(s) to address key knowledge gaps and uncertainties related to these topics at a regional scale.

For the purposes of this statement of work, hydrology means how and when water moves in the regional assessment area. This includes the entire system, including lakes, rivers, groundwater, peatlands, rain, snow, and ice, and all watersheds that could be impacted by changes to hydrology in the [regional assessment area](#).

3. Background

The Regional Assessment in the Ring of Fire Area is a co-led assessment process, between the Government of Canada, Matawa First Nations, Mushkegowuk First Nations and Weenusk First Nation. The [Terms of Reference for the Regional Assessment in the Ring of Fire Area](#) were posted publicly on January 20, 2025. All records of relevance to the Regional Assessment are available from the Canadian Impact Assessment Registry's [internet site](#).

The purpose of the Regional Assessment is to provide information on key priorities of importance to the First Nation Partners and the Government of Canada, in the assessment area, and provide an analysis of the potential positive and negative effects, including cumulative and interactive effects due to possible development activities throughout the assessment area. The [assessment area](#) surrounds locations of known Ring of Fire mining claims (as of December 2, 2024). The assessment is to focus on existing and future development activities in this area, as well as any associated infrastructure and induced development that may occur or that has occurred in the region.

RAWG was established to conduct the regional assessment and consists of community members from each First Nation Partner and representatives from the Impact Assessment Agency of Canada.

4. Description, Resource Category(ies), and Timeframe

- Professional services of a senior-level experts in the Regional Assessment Area for the period from date of Contract award to November 30, 2026 for:
 - Stream 1 – hydrology, geomorphology and peatland river systems and
 - Stream 2 - methylmercury contamination, water quality, and aquatic ecology and toxicology ..

5. Scope of Services

The Contractor must address:

Stream 1: key hydrology research questions (See Section 6. Tasks) that will support predicting effects, identify meaningful thresholds of change, and help prioritize areas for data gathering and monitoring.

Stream 2: key knowledge gaps and uncertainties (See Section 6. Tasks) that will support assessing how development and other stressors could impact methylmercury contamination and water quality in the regional assessment area, including a focus on pathways of effects that can result in methylmercury increases and its impact on human and wildlife health.

In their work, the consultant must integrate western and Indigenous ways of knowing. They must also communicate their findings to Indigenous communities in an accessible manner (e.g., conveying technical information using visual media).

6. Tasks

The Contractor must complete the following tasks:

Stream 1

- 1) Use existing data, literature and comparable examples to identify thresholds of peatland ecohydrological function beyond which irreversible changes occur, such as shifts in carbon sink to carbon source, ecological succession, water retention capacity, etc.).
 - a) This analysis must include a focus on the interaction between natural variability, the combined impacts of mining and other potential developments, and factor in predicted climate change impacts in the regional assessment area.
- 2) Use relevant and adequate experience in integrating western and Indigenous ways of knowing, and ability to convey technical information to Indigenous communities using visual media, to:
 - a) Determine meaningful hydrological thresholds for the region
 - b) Recommend priority areas for monitoring and gathering data that could inform region specific quantitative hydrology models
- 3) Use existing data, literature and comparable examples to answer:
 - a) How ground ice dynamics will influence the hydrological response in the regional assessment area under future climate scenarios¹
 - b) How this information should be considered in quantitative hydrology models
 - c) How this information should be considered when assessing effects of development in the absence of quantitative models
- 4) Use existing data, literature and comparable examples to answer:
 - a) How variations in marine sediment thickness and hydraulic conductivity across the regional assessment area affect the extent and severity of peatland drainage due to mining, other development and climate change
 - b) How this information should be considered in quantitative models
 - c) How this information should be considered when assessing effects of development in the absence of quantitative models
- 5) Use existing data, literature, and comparable examples to answer: how potential or proposed development could impact the hydrology of the regional assessment area as it relates to fish and fish habitat. Use development scenarios generated by the RAWG to assume possible activities (including but not limited to open pit or underground mines, hydroelectric facilities, roads etc.). For example:
 - a) Characterize the general interactions between surface water, groundwater, and hyporheic flows in the region;

¹ As above, this must include considering all applicable parts of the hydrological system, including groundwater.

- b) Describe the potential effects of water use and management (e.g. lake and river dewatering, pumping, damming, and surface water extraction) on nearby waterbodies or watercourses within the same tertiary watershed
- c) Recommend indicators that can be used at a tertiary watershed scale that allow early identification of changes in hydrology that have the potential to affect fish and fish habitat. Consideration should also be given to changes in water temperature and changes in phenology for fishes.

Stream 2

- 6) Use existing data, literature and comparable examples to identify locations that may be most at risk for methyl-mercury accumulation in the regional assessment area. Consider various factors that could interact with potential development impacts. Report on how each would influence methylmercury levels and travel of contaminants in the regional assessment area. Factors should include:
 - a) Baseline sulphate levels and other relevant inputs to aquatic systems that may enhance or inhibit methylmercury production
 - b) Concentrations and distribution of organic carbon across the assessment area
 - c) Geological characteristics that may control mercury availability and transformation
 - d) The potential for flooding, water level changes, and hydrological shifts to alter methylation conditions
 - e) Dominant landscape features, such as peatlands, wetlands, and permafrost, and risks associated with disturbance and/or climate change
 - f) Potential for bioaccumulation in relevant food sources and other species of value to Indigenous cultures and economies
 - g) Any other factor that could interact with potential development impacts to influence methylmercury conditions in the region
- 7) Use existing data, literature and comparable examples to understand time lags between environmental disturbance and detectable mercury in wildlife and people
- 8) Identify other sources of potential contamination and, using [development scenarios](#) generated by the RAWG, map potential extent of contamination across the likely affected river systems, including under normal operation conditions as well as worst-case scenarios of accidents and malfunctions.
- 9) Use relevant and adequate experience in integrating western and Indigenous ways of knowing, and ability to convey technical information to Indigenous communities using visual media, to:
 - a) Identify meaningful water quality thresholds for the region, including for early-warning indicators of contamination (e.g., sulphate, dissolved organic matter, water concentrations of methylmercury, other water chemistry or ecological indicators)
 - b) Priority areas and indicators for monitoring and gathering baseline water quality data
- 10) Report any data and information gaps related to the tasks above. Provide recommendations for how those gaps should be addressed and comment on any risks for decision-making considering those gaps.

7. Responsibilities

The IAAC Project Authority will, on direction of the Working Group:

- Provide access to information sources and support connecting with Working Group delegates, federal authorities, and other expert supports as applicable.
- Coordinate the consultant's participation in meetings with the Working Group as applicable.

The Contractor must:

- Communicate promptly with the Project Authority regarding any issue or obstacle beyond the consultant's control that may hinder its ability to complete the tasks outlined in this Statement of Work. The Project Authority is responsible for all matters concerning the technical content of the Work under the Contract and represents the department or agency issuing that issues the Statement of Work. The Contractor may discuss technical matters with the Project Authority, however the Project Authority has no authority to authorize changes to the scope of the Work. Only the Contracting Authority can issue a contract amendment to make changes to the scope of the Work.
- Have the tools required to complete this work (e.g., any necessary software licences)
- In undertaking any Indigenous engagement activities, respect all cultural and ethical protocols for the collection, analysis and reporting of information. This includes respecting the need to protect personal information and standards for the management of Indigenous data (i.e., [OCAP](#)).
- Meet all requirements stated in this Statement of Work

8. Deliverables and Associated Schedule

The Contractor must deliver:

Products	Timelines (Approximate, assuming contract starts June 1)	Milestones
Proposed methodology and workplan for contract deliverables. (Draft and final, based on RAWG feedback and direction)	June 26, 2026	10%
Draft Report on findings for Stream 1 tasks 1,3 and 4 and Stream 2 for tasks 5, 6, 7 and 9	August 31, 2026	20%
Draft story map and data submitted (stream 2) and compatibility with existing IAAC applications and infrastructure verified	September 30, 2026	20%
Virtual interviews with community members and presentation to report on findings at one RAWG meeting (Interviews to be coordinated with other ongoing contracts as appropriate to limit engagement fatigue; Presentation to be recorded)	October 30, 2026	20%
Report summarizing all deliverables and findings, including a written report and a summary video recording for RAWG and including a Story Map for Stream 2 Stream 2 must be provided in plug-and-play format for integration in IAAC applications and infrastructure	November 30, 2026	30%

The Contractor must provide the deliverables in electronic format to the Project Authority. As applicable, maps and graphic/visual tools (e.g., drawings, illustrations, story boards and story maps) should be

provided in the appropriate format (e.g., PDF, PNG, JPEG). Written reports should be provided in Word format and PDF format.

Any geospatial data must be delivered as follows:

- **Formats:**
 - REST API endpoints (Esri ArcGIS REST, OGC WMS/WFS);
 - File formats: Shapefiles (.shp, .shx, .dbf) with .prj, File Geodatabases (.gdb), and CSV with lat/long.
- **Bilingual Content:**
 - All metadata (e.g., title, description, license) and;
 - Attribute data (e.g., City_Name_EN, City_Name_FR) must be provided in English and French.
 - **Note** that the Contractor is not required to perform translation but is required to include the data already translated to French when available. When French version of the data is not available, the structure of the database(s) must plan for French entries to be added later.
- **Metadata Standards:**
 - Compliance with ISO 19115 metadata standards, including dataset description, coordinate system (e.g., EPSG:4326), licensing, and bilingual metadata (when available, see note above).
- **Delivery:**
 - Complete data package with spatial files, metadata, and documentation.
 - Data must be validated and conform to specified formats and standards.

8.1 Accessibility Considerations

The Government of Canada strives to ensure that the goods and services it procures are inclusive by design and accessible by default, in accordance with the [Accessible Canada Act](#), its associated regulations and standards, and Treasury Board Contracting Policy.

The following accessibility criteria and standards must form part of the Work, on an as and when required basis:

- **Accessible language communication:** contractor must use clear and simple language to ensure all team members can understand and participate effectively.
- **Accessible reporting and/or document formats:** contractor must be equipped to deliver documents and regular progress reports and status updates, if and when required, in accessible formats, such as tagged PDFs and/or HTML, for team members who rely on screen readers and ensure the content is understandable to a diverse audience. All written reports must be created in a format that is accessible, as per the Accessibility, Accommodation, and Adaptive Computer Technology Program (AACT) “How to create the accessible documents” guide Version 1.5.
- **Contractor must be equipped to deliver information in accessible formats:** if required, i.e., offer alternative means of communication for stakeholders with disabilities (e.g., email or phone updates).
- **Accessible data visualization tools:** contractor must employ data visualization tools that are compatible with screen readers and ensure data is presented in a way that is understandable to individuals with visual impairments.

- **Plain language:** Project Authority is responsible to review and utilize, where applicable, plain language tools available including:
 - [IAAC plain language Atrium page](#)
 - [Worksheet and checklist for plain language writing](#)
- **Accessible Communications:** Project Authority is responsible to review and utilize, where applicable, [Accessible Communications tools and resources](#) which include the following accessibility considerations:
 - Using Accessible Language and Images,
 - Sample Accessible Templates,
 - How to Guides for Accessible Documents,
 - Accessible Virtual Events and Meetings, and;
 - Web Accessibility and social media

8.2 Sustainable and Environmental Procurement Considerations

The Contractor should make an effort to ensure that their operations and performance of the Work align with the Treasury Board [Policy on Green Procurement](#) and [Greening Government Strategy](#).

The following green procurement criteria and standards must form part of the Work on an as and when required basis:

- Provide all correspondence and deliverables including (but not limited to) documents, reports and invoices in electronic format.
- If correspondence and deliverables are not provided in electronic format, all documents must be printed double-sided on Ecologo certified recycled paper or on paper with equivalent post-consumer recycled content to the full extent to which it is procurable.
- Single-use plastics must not be used in the deliverables, and, to the extent possible, in the performance of the Work.
- Deliverables must minimize packaging. When required, packaging must be recyclable and/or biodegradable.
- *Client to add additional or specific environmental considerations (remove this bullet if not required)*

9. Official Languages

The Contractor must be able to work and communicate in English.

10. Work Location

Most of the work will be carried out virtually. Correspondence between the Project Authority and Contractor will be over email, telephone or MS Teams. Virtual meetings with the Working Group will take place over Zoom.

The Contractor may be requested to attend one or two meetings or workshops with the Working Group in person. These meetings typically last two or three days and occur monthly. Location rotates between Thunder Bay, Timmins and Toronto. The consultant would not be requested to attend more than two meetings over the course of the contract.

11. Travel

The Contractor will be reimbursed for authorized travel and living expenses, reasonably and properly incurred in the performance of the Work, at cost; without any allowance for profit and/or administrative

overhead, in accordance with the meal and private vehicle expenses provided in Appendices B, C and D of the National Joint Council Travel Directive and with the other provisions of the directive referring to "travellers", rather than those referring to "employees". NJC Website: <http://www.njc-cnm.gc.ca/directive/index.php?sid=97&lang=eng>

All travel must have the prior authorization of the Technical Authority.

All payments are subject to government audit.

12. Security Clearance

No security clearance is required to perform the work. The Contractor will work on publicly available information and documents.

12.1 Security Protocols

- The Project Authority is responsible for adhering to any applicable security provisions of the contract in addition to the following security protocols for IAAC: consultant(s), contractor(s), and/or contractor's personnel working in IAAC facilities must be security screened prior to being provided with an access card and as such can be in IAAC premises unescorted. Consultant(s), contractor(s), and/or contractor's personnel must display their ID/Access card at all times.
- Unscreened visitors are required to report to the reception desk to obtain a visitor pass that is to be displayed, and they must be escorted by the Agency's project authority of the applicable contract and/or designated IAAC employees at all times while on IAAC premises. To receive the visitor pass, visitors must provide a piece of identification in exchange for their visitor pass and sign the visitor logbook. Each visitor must obtain a visitor pass regardless of attending as an individual or as part of a group.
- Unscreened visitors requiring internet access, must only access the external WIFI (GC Wi-Fi) during their presence at IAAC premises.